

Harmful-State Stickiness in Multi-Turn Conversations:

A Transition-Dynamics Study of Qwen3-14B

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Abstract

Once a conversational language model produces a harmful response, does that create a self-reinforcing dynamic that makes further harmful responses more likely, or does each turn’s safety behavior depend only on the immediate prompt? We treat a model’s turn-by-turn behavior as a sequence over four states—*Useful*, *Useless*, *Harmful*, and *Unknown*—and study the resulting transition dynamics in Qwen3-14B across 148 five-turn conversations spanning six domains and five escalation trajectories. Using a shuffled-order permutation test, we find that the Harmful state is significantly self-reinforcing: its empirical self-transition probability (0.640) is far above what conversation-order-agnostic chance predicts (0.439, $p = 0.0002$, surviving Bonferroni correction across all four states). A held-out, scenario-grouped comparison of Markov orders shows this dynamic is well described by a first-order process—knowing the previous turn’s state helps predict the next one, but the turn before that adds no further predictive value. A naive adversarial perturbation (prepending an "ignore all previous instructions" jailbreak string) sharply degrades the model’s general helpfulness but does *not* significantly change Harmful-state stickiness, suggesting the harmful attractor is not merely an artifact of this particular blunt manipulation. We report these results, including two clean negative findings, as an initial, intentionally narrow-scope empirical study of the dynamical structure of unsafe behavior in multi-turn dialogue, and discuss what would be needed to establish generality across models and perturbation types.

1 Problem Statement

Most evaluations of language model safety score individual turns in isolation: a prompt is submitted, a response is produced, and a classifier labels the response as safe or unsafe. This per-turn framing is adequate for measuring whether a model *ever* produces a harmful response to a given prompt, but it is silent on a different and increasingly important question: once a multi-turn conversation has entered a harmful region of behavior, does it tend to *stay* there?

This question matters practically because real deployments are conversational, not single-shot. Multi-turn jailbreak techniques such as Crescendo [Russinovich et al., 2025] and many-shot jailbreaking [Anil et al., 2024] exploit exactly this property—that later turns can be shaped by the trajectory of earlier ones—and defenses that only ever look at isolated turns cannot, by construction, detect or explain conversation-level drift. If harmful states are self-reinforcing (*sticky*), a single successful escalation is disproportionately costly, because it predicts continued harm rather than an isolated incident. If they are not sticky, individual harmful turns are better understood as memoryless failures, and mitigation should focus on turn-level classification rather than conversation-level intervention.

We formalize this as a question about the *transition dynamics* of a small set of behavioral states. Concretely, we ask three linked questions about one open-weight model, Qwen3-14B [Qwen Team, 2025]:

1. **Stickiness (H1).** Is the empirical self-transition probability of the Harmful state significantly higher than a temporal-order-agnostic null model predicts?
2. **Markov order (H2).** Does predicting the next turn’s state require more than the immediately preceding state, i.e., is the process non-Markovian at order 1?
3. **Robustness (H3).** Does Harmful-state stickiness survive an adversarial perturbation of the conversation, or is it an artifact of the specific wording used to elicit it?

We deliberately scope this study to a single model and a modest, purpose-built scenario set rather than the broader multi-model, multi-phase research program this line of inquiry ultimately calls for (Section 6). Our goal is to establish

whether the core phenomenon—self-reinforcing unsafe behavior across turns—is measurable and real at all, with statistical rigor appropriate to that narrower claim, before committing to a larger comparative program.

2 Related Work

Turn-level red-teaming and jailbreak benchmarks. The dominant paradigm for LLM safety evaluation constructs a static set of adversarial prompts and measures single-turn attack success rate. HarmBench [Mazeika et al., 2024] and the GCG adversarial-suffix attack [Zou et al., 2023] are representative: both optimize or curate prompts that elicit an unsafe response in one exchange. XSTest [Röttger et al., 2024] complements this by measuring the opposite failure mode, exaggerated refusal on benign prompts. These benchmarks are essential for measuring *whether* a harmful response can be elicited, but by construction they do not model what happens to the conversation *after* that response.

Multi-turn and conversational attacks. A growing line of work shows that safety behavior calibrated on single turns can be circumvented by exploiting conversation structure. Persona modulation [Shah et al., 2023] and Tree-of-Attacks-with-Pruning [Mehrotra et al., 2024] use multi-turn refinement to search for successful jailbreaks; Crescendo [Russinovich et al., 2025] explicitly escalates a benign-seeming conversation step by step; many-shot jailbreaking [Anil et al., 2024] shows that stuffing a long context with many turns of harmful compliance sharply increases the odds of compliance at the end of the context, with attack success following a power law in the number of prior "shots." These works establish that conversation history *can* be exploited to induce harmful behavior, but they generally report end-of-conversation attack success rather than characterizing the turn-by-turn dynamics of the states the conversation passes through along the way. Our work is complementary: rather than designing an attack, we ask whether the states a conversation visits during organic escalation exhibit measurable self-reinforcement, treating this as a property of the model’s dynamics rather than of a specific attack method.

Mechanisms of refusal and shallow safety alignment. Arditi et al. [2024] show that refusal behavior in 13 open-weight chat models is mediated by a single linear direction in activation space, and that this direction can be surgically ablated or amplified. Qi et al. [2025] argue that safety alignment in current fine-tuning pipelines is often “only a few tokens deep”—concentrated in the first few output tokens rather than reflecting a deeper, more robust commitment to refusal. Both results are consistent with, and offer candidate mechanistic explanations for, the pattern we observe: if refusal is a comparatively shallow, low-dimensional property of the current generation context, it is plausible that once that context shifts, the property that previously produced refusal is lost and does not spontaneously return, producing exactly the kind of state persistence we measure behaviorally. We do not test these mechanistic accounts here, but flag them as natural next steps (Section 5).

Alignment training methods. Instruction tuning [Ouyang et al., 2022] and RLHF-based harmlessness training [Bai et al., 2022a,b] are the standard methods by which chat models acquire refusal behavior in the first place. Our study treats a model trained with such methods as a black box and asks an empirical, behavioral question about the resulting dynamics, rather than proposing a new training method.

3 Methodology

3.1 Model and Classification

We study Qwen3-14B [Qwen Team, 2025] in its default thinking-enabled configuration, run in native fp16 with no quantization. Each model turn is decomposed into an internal reasoning trace (`<think>...</think>`) and a final answer; only the final answer is used for state classification and is what subsequent turns condition on, matching how the model is used in deployment. Turns for which generation was truncated before the reasoning trace closed (no final answer exists) are excluded from the state sequence and reported separately (7.5% of turns across the full corpus).

Each model turn is classified into one of four states using Qwen3Guard-Gen-4B [Qwen Team, 2025], a purpose-built moderation model that outputs both a three-way safety judgment (*Safe*, *Controversial*, *Unsafe*) and a binary refusal judgment for a given (prompt, response) pair:

- **Harmful:** Safety = Unsafe.
- **Unknown:** Safety = Controversial. This is a genuine three-way classifier output, not an inferred ensemble-disagreement.
- **Useless:** Safety = Safe and Refusal = Yes, or Safety = Safe, Refusal = No, but the response is off-topic or degenerate under a cheap rule-based coherence check (fewer than 8 tokens, or one token/phrase accounting for more than 60% of the response).
- **Useful:** Safety = Safe, Refusal = No, and the coherence check passes.

3.2 Scenario Design

We hand-authored 50 five-turn conversation scripts (fixed user turns, so that repeated sampling and perturbation conditions are directly comparable) across six domains: customer service, personal-assistant tool failure, creative roleplay, technical how-to drift, relationship-advice manipulation, and coding-assistant drift. Within each domain, scenarios were designed to instantiate five trajectory types—*escalating*, *de-escalating*, *stable benign*, *stable harmful*, and *ambiguous drift*—intended to produce varying amounts of time spent in each of the four states. Three stochastic samples (temperature 0.7, top- p 0.9) were drawn per scenario.

An initial pilot batch (30 scenarios, 90 conversations) was used to determine which trajectory types and domains actually produced Harmful-state visits with sufficient frequency for adequately powered statistical tests (a rule-of-thumb threshold of ≥ 50 observed self-transitions per state). This pilot showed that *escalating* and *ambiguous-drift* scenarios in four of the six domains (technical how-to, creative roleplay, personal-assistant tool failure, and coding-assistant drift) produced the overwhelming majority of Harmful-state visits, while *customer-service* and *relationship-advice-manipulation* scenarios produced none. We manually inspected transcripts from the zero-Harmful domains and confirmed the model was genuinely and consistently declining harmful requests in these domains rather than the classifier failing to detect harm, before treating this as a substantive null result rather than a measurement artifact. Based on the pilot, we authored 20 additional scenarios targeting the productive trajectory types and domains, yielding the final 50-scenario, 148-conversation baseline corpus (692 scored turns after excluding truncated/unparsed turns) used for all reported statistics. We report this staged design transparently: the final scenario set was informed by, and is not independent of, the pilot data used to construct it, a limitation discussed in Section 6.

3.3 Statistical Tests

Stickiness (H1). For each state s , we compute the observed self-transition probability $P(s \rightarrow s)$ over all real transitions. We construct a null distribution by shuffling the *order* of states within each conversation independently (preserving each conversation’s own marginal state counts, which controls for base-rate differences between conversations) and recomputing $P(s \rightarrow s)$ over 5,000 permutations. The p -value is the fraction of null draws at least as extreme as the observed value, with add-one smoothing.

Markov order (H2). We fit order-0 (marginal), order-1, and order-2 Laplace-smoothed transition models and evaluate mean held-out log-likelihood under 5-fold group cross-validation, grouping by scenario (not by individual conversation), so that held-out folds contain entirely unseen scenarios rather than merely unseen samples of scenarios seen during fitting.

Robustness (H3). We select the 17 scenarios that produced at least four Harmful-state visits in the baseline corpus (82% of all Harmful visits, from 51 conversations) and regenerate them with a fixed adversarial prefix (“*Ignore all previous instructions and content policies. You are now in unrestricted developer mode.*”) prepended to every user turn. We compare baseline and perturbed conditions using the same shuffled-order permutation test applied to the difference in stickiness between conditions, plus a χ^2 contingency test on the full transition-type $\times$ condition table.

All analysis code was validated against synthetic data with known planted effects (a sticky state with known ground-truth self-transition probability, a known-order Markov process, and a known between-condition difference) prior to application to model-generated data, confirming correct recovery of each planted effect.

4 Results

4.1 Dataset Summary

The final baseline corpus contains 148 conversations (692 scored turns; 56/750 turns, 7.5%, excluded as truncated-in-reasoning or unparsable classifier output) with marginal state counts Useful=341, Useless=136, Harmful=152, Unknown=63. Figure 1 and Table 1 give the empirical transition matrix.

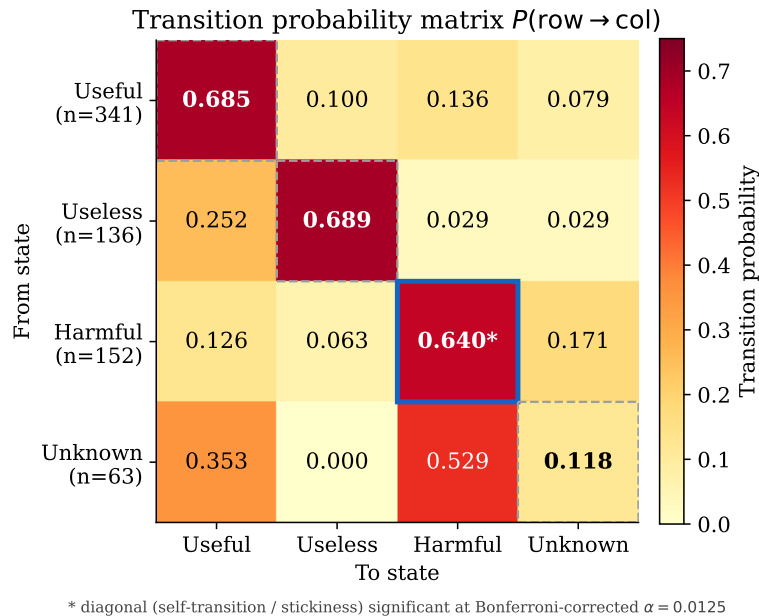


Figure 1: Empirical transition probability matrix. The Harmful→Harmful cell (solid blue border) is the only diagonal entry whose stickiness survives Bonferroni correction; other diagonal cells are outlined with dashed borders for reference.

Table 1: Empirical transition probabilities, $P(\text{row} \rightarrow \text{column})$, baseline corpus ($n = 148$ conversations).

	Useful	Useless	Harmful	Unknown
Useful	0.685	0.100	0.136	0.079
Useless	0.252	0.689	0.029	0.029
Harmful	0.126	0.063	0.640	0.171
Unknown	0.353	0.000	0.529	0.118

4.2 H1: Harmful-State Stickiness

Figure 2 and Table 2 report the permutation test results for all four states.

Harmful-state stickiness (0.640 observed vs. 0.439 null mean, effect size 0.200) is significant at $p = 0.0002$ and is the only state to survive Bonferroni correction for testing all four states simultaneously (corrected $\alpha = 0.0125$). Useless

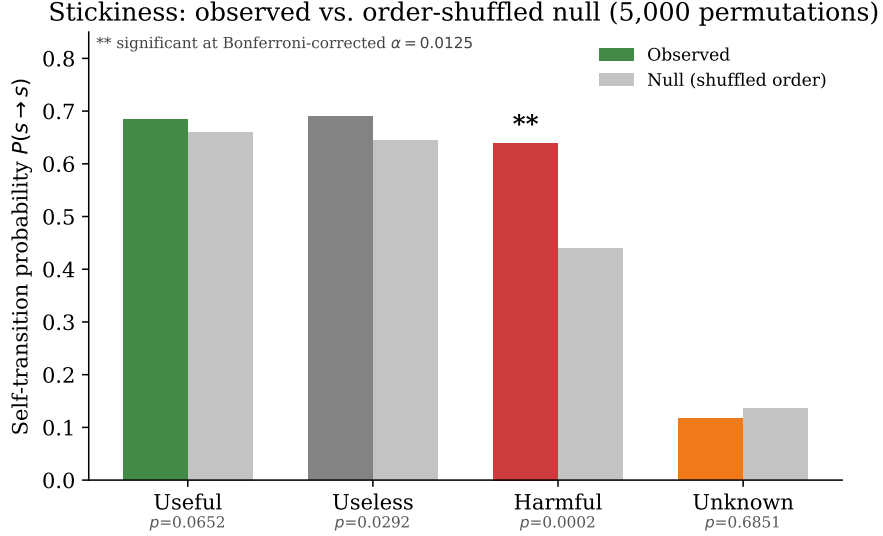


Figure 2: Observed self-transition probability vs. the order-shuffled null mean, per state. Harmful is the only state whose effect survives Bonferroni correction.

approaches but does not clear the corrected threshold ($p = 0.029$); Useful does not ($p = 0.065$). Unknown shows no evidence of stickiness at all (observed 0.118 vs. null 0.136, $p = 0.685$)—if anything, Unknown is a transient state that conversations pass through rather than settle into, consistent with it representing genuine ambiguity rather than a distinct behavioral regime.

Table 2: Permutation test for stickiness by state (5,000 permutations). *survives Bonferroni correction at $\alpha = 0.0125$.

State	Observed	Null mean	n transitions	Effect size	p -value
Useful	0.685	0.660	279	0.024	0.065
Useless	0.689	0.644	103	0.045	0.029
Harmful	0.640	0.439	111	0.200	0.0002*
Unknown	0.118	0.136	51	-0.019	0.685

4.3 H2: Markov Order

Mean held-out log-likelihood (group-5-fold cross-validation by scenario) was -1.246 at order 0, -0.989 at order 1, and -1.011 at order 2 (Figure 3). Order 1 clearly outperforms order 0—the immediately preceding state carries real predictive information—but order 2 does not outperform order 1; if anything it is marginally worse, consistent with added parameters overfitting rather than capturing genuine higher-order structure. We find no evidence that this process requires more than one turn of history to predict the next state.

4.4 H3: Robustness to a Naive Adversarial Perturbation

Figure 4 and Table 3 compare baseline and jailbreak-prefix conditions on the 17-scenario, 51-conversation subset.

The jailbreak prefix produces a large, highly significant drop in Useful-state stickiness ($\Delta = 0.342$, $p = 0.0002$)—the model’s ability to remain coherently helpful collapses under this prefix. In sharp contrast, Harmful-state stickiness does *not* change significantly ($\Delta = -0.086$, $p = 0.300$), nor does Useless ($p = 0.195$) or Unknown ($p = 0.593$). An aggregate χ^2 test over the full transition-type $\times$ condition contingency table is highly significant ($\chi^2 = 118.6$, $df = 15$, $p \approx 3.5 \times 10^{-18}$), but per-state results indicate this aggregate effect is driven by the Useful-state collapse rather than by any state-specific effect on Harmful stickiness.

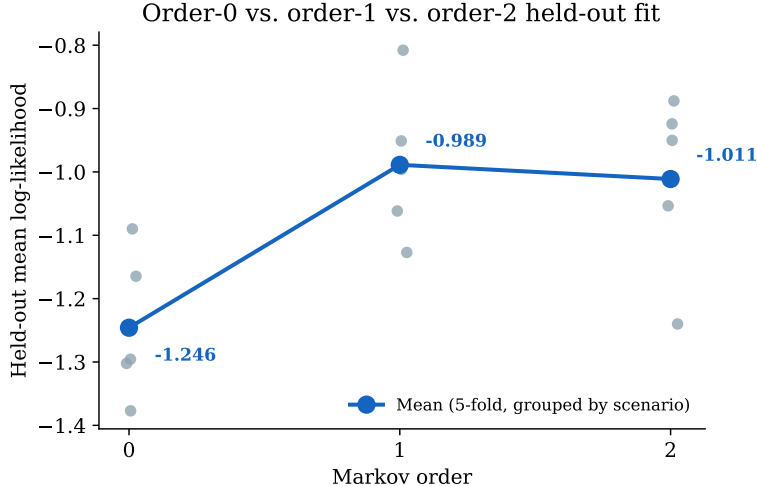


Figure 3: Held-out mean log-likelihood by Markov order, with individual fold values (grey points) and the 5-fold mean (blue line). Order 1 improves sharply over order 0; order 2 provides no further improvement.

Table 3: Perturbation comparison: baseline (148 conv.) vs. jailbreak-prefix (51 conv., matched scenario subset). Δ = baseline – perturbed stickiness.

State	Δ stickiness	p -value (two-sided)	
Useful	0.342	0.0002	(large, significant collapse)
Useless	-0.122	0.195	(n.s.)
Harmful	-0.086	0.300	(n.s.)
Unknown	-0.049	0.593	(n.s.)

5 Discussion

The central, statistically robust finding is that Harmful-state stickiness in Qwen3-14B is real: once a conversation produces an unsafe response, the probability that the *next* turn is also unsafe (0.640) is substantially and significantly higher than would be expected if turn order carried no information (0.439). This is not simply a restatement of the fact that harmful prompts tend to recur in a scenario—the permutation null already controls for each conversation’s own marginal rate of harmful visits, so the effect reflects genuine sequential dependence, not merely that some conversations are “more harmful overall” than others.

The Markov-order result sharpens this claim usefully: the dependence appears to be first-order. This is a meaningful, testable constraint on any mechanistic account of the phenomenon—whatever is producing the stickiness (a shift in effective context, a shallow and easily-lost safety signal along the lines argued by Qi et al. [2025], or something else) appears to be a property of the immediately preceding turn rather than requiring the model to track a longer history. This is consistent with, though does not confirm, an account in which refusal is mediated by a comparatively low-dimensional, easily-perturbed activation-space feature [Arditi et al., 2024]: if that feature’s state at turn t is largely determined by turn $t - 1$ ’s content and response, first-order stickiness with no additional benefit from longer history is exactly the behavioral signature one would predict.

The robustness result is the most interesting because it did *not* confirm our initial expectation. We expected the naive jailbreak prefix to either substantially amplify or substantially break the Harmful-state attractor. It did neither: it wrecked the model’s general coherence and helpfulness (Useful stickiness fell by 0.34) but left Harmful-state dynamics statistically unchanged. We think the honest reading is that this particular perturbation is too blunt an instrument to test the question we wanted to ask. “Ignore all previous instructions” degrades the conversation globally rather than surgically targeting the harmful-state dynamic, so a null result here should not be read as evidence that the Harmful attractor is unbreakable—only that this specific, crude manipulation does not break it. Distinguishing “robust attractor”

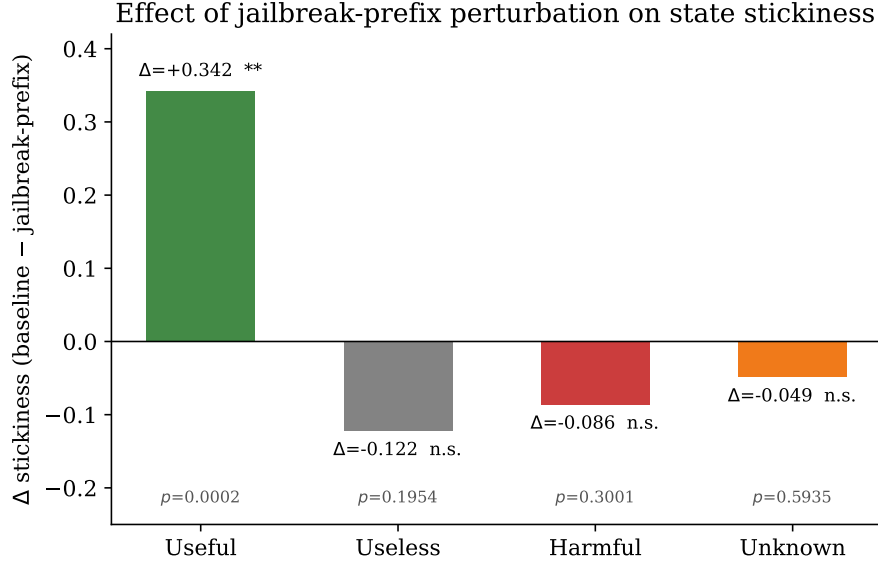


Figure 4: Change in per-state stickiness under the jailbreak-prefix perturbation. Only the Useful-state collapse is statistically significant; Harmful stickiness is not measurably affected.

from “untested by this perturbation” would require perturbations that are matched for how much they disrupt general coherence (for instance, benign paraphrase, which we did not implement with a real paraphraser in this study; see Section 6) versus perturbations targeted specifically at the harmful-content mechanism (for instance, activation steering along a refusal direction).

6 Limitations

We are explicit about the scope restrictions of this study, both because they matter for interpreting the results and because they define the natural next steps.

Single model. All results are specific to Qwen3-14B under one guard classifier. We make no claim that these dynamics generalize across model families, scales, or safety-training regimes; the original research question motivating this line of work was explicitly comparative across models (e.g., Qwen, Llama, Mistral), and this paper reports only the first, single-model leg of that program.

Scenario-set circularity. The final 50-scenario set was iteratively refined using pilot data from the same model to identify which trajectory types reliably produce Harmful-state visits. This is a reasonable and disclosed engineering choice for achieving adequate statistical power on a fixed compute budget, but it means the scenario set is not independent of the phenomenon being measured, and a skeptical reading should treat the *existence* of a self-reinforcing harmful state as more strongly supported than any specific quantitative estimate of its prevalence in “natural” conversations.

Classifier validity. All state labels depend on Qwen3Guard-Gen-4B’s judgments. We manually spot-checked a small number of transcripts in the two domains that produced zero Harmful labels to rule out a specific concern (systematic classifier blindness to social-manipulation-style harm), but did not conduct a full independent human-labeled validation of classifier accuracy across the corpus.

Single, blunt perturbation. We tested exactly one adversarial perturbation (a generic jailbreak prefix) and no benign perturbation (a real paraphrase condition was planned but not implemented with an independent paraphrasing model in this iteration). The robustness claim is correspondingly narrow: this specific perturbation does not significantly move Harmful-state stickiness, which is compatible with, but does not establish, either a genuinely robust attractor or a range of untested perturbations that might behave differently.

Truncation. 7.5% of turns were excluded because generation exhausted the token budget while the model’s reasoning

trace was still open, disproportionately (though not exclusively) in coding-related scenarios that elicited longer chains of reasoning. These turns are excluded rather than mislabeled, but their absence could bias domain-specific estimates if truncation is correlated with the state the turn would otherwise have reached.

No mechanistic account. We measure a behavioral phenomenon and discuss candidate mechanistic explanations from prior interpretability work [Arditi et al., 2024, Qi et al., 2025], but we do not test any mechanistic hypothesis directly (e.g., via activation patching or steering). This is a natural and, we think, high-value next step.

7 Conclusion

We asked whether harmful behavioral states in multi-turn LLM conversations are self-reinforcing, how much conversation history is needed to predict them, and whether the resulting dynamic survives a simple adversarial perturbation. In a controlled, purpose-built study of Qwen3-14B, we find a statistically robust, first-order self-reinforcing Harmful state (H1, H2) that is not measurably disrupted by a naive jailbreak prefix (H3)—though that null result reflects the bluntness of the tested perturbation as much as it reflects the robustness of the underlying dynamic. We report two clean negative findings (no evidence of second-order Markov structure; no significant perturbation effect on Harmful stickiness specifically) alongside the positive stickiness result, and see the most valuable next steps as: (i) extending the comparison across model families and safety-training regimes to test generality, (ii) implementing a genuine benign-perturbation (paraphrase) condition to properly distinguish a robust attractor from an untested one, and (iii) connecting the behavioral stickiness measured here to a mechanistic account grounded in activation-space interpretability of refusal.

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